21 NOV 2025

25FEBS101ESE25A

Seat No. : .....

Student Name : .....

St. John College of Engineering and Management, Palghar, Maharashtra

Autonomous Institute

End Semester Examinations, November 2025

ASH - SEMESTER I

25FEBS101 - Engineering Mathematics – I

Time : 2.:00 Hours **Maximum Marks : 60**

St. John College of Engineering & Management ★ SJCEM PALGHAR

Instructions

- All Questions are compulsory
- Assume suitable data wherever required

		Attempt all Questions	Mrk	Course Outcome	RBT Level
1.	a	i. Find the rank of $A = \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -2 & 1 \\ 1 & -1 & 4 & 0 \\ -2 & 2 & 8 & 0 \end{pmatrix}$ by reducing to echelon form.	5	CO1	L3
		ii. Show that $A = \frac{1}{9}\begin{pmatrix} -8 & 4 & 1 \\ 1 & 4 & -8 \\ 4 & 7 & 4 \end{pmatrix}$ is orthogonal.	5	CO1	L3
		OR			
	b	Given a system of linear equations with parameters λ and μ, evaluate and justify for which values of λ and μ the system has (i) no solution (ii) a unique solution (iii) infinite number of solutions. $2x + 3y + 5z = 9; 7x + 3y - 2z = 8; 2x + 3y + \lambda z = \mu$	10	CO1	L5
2.	a	i. If $u = tan^{-1}(\frac{y}{x})$, show $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$	5	CO2	L3
		ii. If $u = x^2y$ and $x^3 + y^3 + 3xy = 0$, find $\frac{du}{dx}$.	5	CO2	L3
		OR			
	b	**Evaluate** the validity of the identity $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = (l^2 + m^2)\left(\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2}\right)$ if z is a function of x and y and u, v are two other variables such that $u = lx + my, y = ly - mx$.	10	CO2	L5
3.	a	i. Discuss the maximum and minimum of $x^3 + y^3 - 3axy, a > 0$.	5	CO3	L3
		ii. If $u = sin^{-1}\left(\frac{x^{\frac{1}{4}} + y^{\frac{1}{4}}}{x^{\frac{1}{5}} + y^{\frac{1}{5}}}\right)$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{20}tan(u)$	5	CO3	L3

		OR			
	b	**Evaluate** the validity of using Lagrange multipliers to identify the maximum and minimum values of $f(x,y) = x^2 + y^2$ subject to the constraint $x + y = 2$.	10	CO3	L5
4.	a	i. **Analyze** the given **indeterminate form** to identify the suitable method for evaluation, and then **evaluate** $lim_{x\to\infty} \frac{log(x+\sqrt{x^2+1})}{log(x+\sqrt{x^2-1})}$	5	CO4	L4
		ii. **Apply** the **Maclaurin's series** to **show that** $sec\, x = 1 + \frac{x^2}{2} + \frac{5x^4}{24} + \ldots$	5	CO4	L3
		OR			
	b	**Evaluate** the given function by applying **Taylor's Theorem**, and **show that** $\sqrt{1+x+2x^2} = 1 + \frac{x}{2} + \frac{7}{8}x^2 - \frac{7}{16}x^3 + \ldots$	10	CO4	L5
5.	a	i. **Apply** the definition and properties of the **Beta function** to **prove that** $\int_0^{\pi} \sin^2\theta(1+\cos\theta)^3\, d\theta = \frac{7\pi}{8}$	5	CO5	L4
		ii. **Apply** the properties of the **Gamma function** to **evaluate** $\int_0^{\infty} \frac{x^4}{4^x}\, dx$	5	CO5	L3
		OR			
	b	**Evaluate** the validity of the given result by applying the method of **Differentiation Under Integral Sign (DUIS)**. $\int_0^{\infty} \frac{1-\cos mx}{x} e^{-x}\, dx = \frac{1}{2}log(m^2+1)$	10	CO5	L5
6.	a	i. **Apply** the **Newton–Raphson method** to find the **root of the given equation** $x^3 - 9x^2 - 18 = 0$, correct up to **four decimal places**.	5	CO6	L3
		ii. **Apply** the **False Position (Regula Falsi) method** to find the **approximate root** of the given equation $2x - 3sinx - 5 = 0$, performing **three iterations**.	5	CO6	L3
		OR			
	b	**Analyze** the given system of linear equations and apply the **Gauss–Jacobi iterative method** to determine the approximate solution up to **four iterations**. $15x + 2y + z = 18, 2x + 20y - 3z = 19, 3x - 6y + 25z = 22$	10	CO6	L4

* All the Best *

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